Further Kinematics

Solutions

Overview

Solution 2

Use vectors to represent motion. Can we use the *suvat* equations with constant acceleration? What changes when acceleration varies with time t?

1. Vector equations for motion

A particle has velocity

$\mathbf{v} = (1 - 2t)\mathbf{i} + (3t - 3)\mathbf{j} \quad \text{m s}^{-1}$.

Find (a) its speed at $t = 0$; (b) its bearing at $t = 2$; (c) when it moves (i) parallel to $\mathbf{j}$, (ii) parallel to $-\mathbf{i} - 3\mathbf{j}$. (3, 2, 6 marks)

2. Variable acceleration with vectors

A particle P, mass $0.8\,\text{kg}$, is acted on by a single force $\mathbf{F}$. Its position relative to fixed origin O is

$\mathbf{r} = 2t^3\mathbf{i} + 50t^{-1/2}\mathbf{j} \quad \text{m}, \quad t > 0$.

Find (a) its speed at $t = 4$; (b) acceleration as a vector at $t = 2$; (c) $\mathbf{F}$ at $t = 2$.

3. Integration for velocity/displacement

A particle moving in a plane has velocity

$\mathbf{v} = 3t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} \quad \text{m s}^{-1}, \quad t \geq 0$.

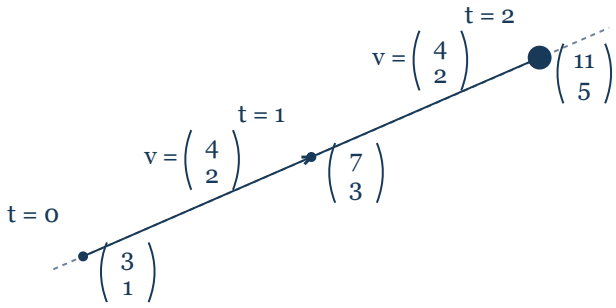
At $t = 0$, its position relative to O is $(2\mathbf{i} - 3\mathbf{j})\,\text{m}$. Find its position vector at time t.

Vector Motion

Solution 3

Initially, a ball has position vector $\begin{pmatrix}3\\1\end{pmatrix}$ m. Each second it moves by $\begin{pmatrix}4\\2\end{pmatrix}$ m: its constant velocity is $\begin{pmatrix}4\\2\end{pmatrix}\,\mathrm{m\,s^{-1}}$.

Where is it after 1 second? After 2 seconds?



1 second

After 1 second:

$$\begin{pmatrix}3\\1\end{pmatrix}+\begin{pmatrix}4\\2\end{pmatrix}=\begin{pmatrix}7\\3\end{pmatrix}\text{ m.}$$

2 seconds

After 2 seconds:

$$\begin{pmatrix}3\\1\end{pmatrix}+2\begin{pmatrix}4\\2\end{pmatrix}=\begin{pmatrix}11\\5\end{pmatrix}\text{ m.}$$

In general, where is the ball after t seconds?

General

Add t lots of $\begin{pmatrix}4\\2\end{pmatrix}$ to $\begin{pmatrix}3\\1\end{pmatrix}$:

$$\begin{pmatrix}3\\1\end{pmatrix}+t\begin{pmatrix}4\\2\end{pmatrix}=\begin{pmatrix}3+4t\\1+2t\end{pmatrix}.$$

Position vector

$\mathbf{r}=\mathbf{r}_0+\mathbf{v}t$,

where $\mathbf{r}_0$ is initial position and $\mathbf{v}$ is constant velocity.

Example

Solution 4

A particle starts at $(3\mathbf{i}+7\mathbf{j})$ m, moving with constant velocity $(2\mathbf{i}-\mathbf{j})\,\mathrm{m\,s^{-1}}$.

(a) Find its position vector 4 seconds later.

(b) Find when it is due east of the origin.

a

$$\mathbf{r}=\begin{pmatrix}3\\7\end{pmatrix}+4\begin{pmatrix}2\\-1\end{pmatrix}=\begin{pmatrix}11\\3\end{pmatrix}\text{ m.}$$

b

Due east means the **j** component is zero and the **i** component is positive:

$$\mathbf{r}=\begin{pmatrix}3\\7\end{pmatrix}+t\begin{pmatrix}2\\-1\end{pmatrix}=\begin{pmatrix}3+2t\\7-t\end{pmatrix}.$$

$7-t=0 \quad\Rightarrow\quad t=7\,\mathrm{s}.$

The east component is then $17\,\mathrm{m}>0$.

Column vectors may be easier to manipulate than i and j notation; either may be used.

SUVAT with Vectors

Solution 5

Some *suvat* equations work with vectors. Here $\mathbf{r}$ denotes displacement from the initial position:

$\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2, \qquad \mathbf{v} = \mathbf{u} + \mathbf{a}t.$

Which quantities are vectors, and which are scalars?

r **r**: vector.

u **u**: vector.

v **v**: vector.

a **a**: vector.

t: scalar. The scalar formula $v^2 = u^2 + 2as$ cannot be used by simply squaring vector symbols.

A particle has initial velocity $(-3\mathbf{i} + \mathbf{j})\,\mathrm{m\,s^{-1}}$ and constant acceleration $(2\mathbf{i} + 3\mathbf{j})\,\mathrm{m\,s^{-2}}$. Find (a) its speed and (b) its bearing at $t = 3\,\mathrm{s}$.

Speed

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$= \begin{pmatrix} -3 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 10 \end{pmatrix} \mathrm{m\,s^{-1}}.$

Speed $= \sqrt{3^2 + 10^2} = 10.4\,\mathrm{m\,s^{-1}}$ (3 s.f.).

Speed is the scalar magnitude of velocity.

Bearing



$\tan\theta = \frac{3}{10} \Rightarrow \theta = 16.7°.$

The bearing is 017°. The velocity vector gives the direction of motion; draw it to establish the angle.

Further Example

Solution 6

An ice skater is at fixed point O at $t = 0$, with velocity $(2.4\mathbf{i} - 0.6\mathbf{j})\,\mathrm{m\,s^{-1}}$. At $t = 20\,\mathrm{s}$, velocity is $(-5.6\mathbf{i} + 3.4\mathbf{j})\,\mathrm{m\,s^{-1}}$. Model the skater as a particle with constant acceleration. Find (a) acceleration; (b) position $\mathbf{s}$ relative to O at time t; (c) when directly north-east of O. A second skater has position $\mathbf{r} = (1.1t - 6)\mathbf{j}\,\mathrm{m}$. (d) Show that they meet.

a

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$\begin{pmatrix} -5.6 \\ 3.4 \end{pmatrix} = \begin{pmatrix} 2.4 \\ -0.6 \end{pmatrix} + 20\mathbf{a}$

$20\mathbf{a} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$

$\mathbf{a} = \begin{pmatrix} -0.4 \\ 0.2 \end{pmatrix} \mathrm{m\,s^{-2}}.$

b

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$= \begin{pmatrix} 2.4 \\ -0.6 \end{pmatrix} t + \frac{1}{2}\begin{pmatrix} -0.4 \\ 0.2 \end{pmatrix} t^2$

$= \begin{pmatrix} 2.4t - 0.2t^2 \\ -0.6t + 0.1t^2 \end{pmatrix} \mathrm{m}.$

c

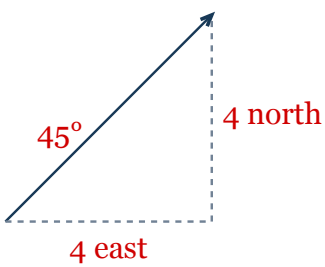
North-east: equal positive components.

$2.4t - 0.2t^2 = -0.6t + 0.1t^2$

$3t(1 - 0.1t) = 0$

$t = 0 \quad \text{or} \quad t = 10.$

Reject $t = 0$, at the origin. At $t = 10$, both components are 4, so the time is 10 s.



d

Meeting requires equal position vectors:

$\begin{pmatrix} 2.4t - 0.2t^2 \\ -0.6t + 0.1t^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1.1t - 6 \end{pmatrix}.$

$2.4t - 0.2t^2 = 0$

$0.2t(12 - t) = 0 \Rightarrow t = 0 \text{ or } 12.$

At $t = 0$, north components are 0 and -6: no meeting. At $t = 12$:

$-0.6(12) + 0.1(12^2) = 7.2,$

$1.1(12) - 6 = 7.2.$

They meet at $t = 12\,\mathrm{s}$.

Test Your Understanding — Solution 7

Edexcel M1 (Old), May 2013(R), Q6

Unit vectors $\mathbf{i}, \mathbf{j}$ point east and north. Ship S has constant velocity $(3\mathbf{i} + 3\mathbf{j})\,\text{km h}^{-1}$; at $t = 0$, position is $(-4\mathbf{i} + 2\mathbf{j})$ km. (a) Find its position at time t hours. Ship T has velocity $(-2\mathbf{i} + n\mathbf{j})\,\text{km h}^{-1}$, initial position $(6\mathbf{i} + \mathbf{j})$ km. They meet at P. (b) Find n. (c) Find OP. (2,5,4 marks)

a

$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

$= (-4\mathbf{i} + 2\mathbf{j}) + (3\mathbf{i} + 3\mathbf{j})t$

$= (-4 + 3t)\mathbf{i} + (2 + 3t)\mathbf{j}.$

b

For ship T:

$\mathbf{r}_T = (6\mathbf{i} + \mathbf{j}) + (-2\mathbf{i} + n\mathbf{j})t$

$= (6 - 2t)\mathbf{i} + (1 + nt)\mathbf{j}.$

At meeting, positions are identical:

$-4 + 3t = 6 - 2t$

$5t = 10 \Rightarrow t = 2.$

$2 + 3(2) = 1 + 2n \Rightarrow n = 3.5.$

c

At $t = 2$, position of P:

$(-4 + 6)\mathbf{i} + (2 + 6)\mathbf{j} = 2\mathbf{i} + 8\mathbf{j}.$

$OP = \sqrt{2^2 + 8^2}$

$= \sqrt{68} = 8.25\,\text{km}$

(3 s.f.).

Exercise 8A — Y2 Mechanics Pages 162–164 — Solution 8

10. A model boat has constant acceleration $(0.2\mathbf{i} + 0.6\mathbf{j})\,\text{m s}^{-2}$. After 20 seconds its velocity is $(4\mathbf{i} + 3\mathbf{j})\,\text{m s}^{-1}$. Find displacement from its starting position after 20 seconds.

Working

$\mathbf{v} = 4\mathbf{i} + 3\mathbf{j},\ t = 20,\ \mathbf{a} = 0.2\mathbf{i} + 0.6\mathbf{j}$. Find $\mathbf{s}$:

$\mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$

$= 20(4\mathbf{i} + 3\mathbf{j}) - \frac{20^2}{2}(0.2\mathbf{i} + 0.6\mathbf{j})$

$= 80\mathbf{i} + 60\mathbf{j} - 40\mathbf{i} - 120\mathbf{j}$

$= 40\mathbf{i} - 60\mathbf{j}.$

After 20 seconds, displacement is $(40\mathbf{i} - 60\mathbf{j})$ m.

Exercise 8A — Y2 Mechanics Pages 162–164 Solution 9

11. Particle A starts at $(12\mathbf{i}+12\mathbf{j})$ m, with velocity $(-\mathbf{i}+\mathbf{j})\,\mathrm{m\,s^{-1}}$ and constant acceleration $(2\mathbf{i}-4\mathbf{j})\,\mathrm{m\,s^{-2}}$. Particle B has initial velocity $\mathbf{i}\,\mathrm{m\,s^{-1}}$ and constant acceleration $2\mathbf{j}\,\mathrm{m\,s^{-2}}$. They collide after 3 seconds. Find (a) both speeds then; (b) the collision point's position vector; (c) B's initial position vector.

a

Use $\mathbf{v}=\mathbf{u}+\mathbf{a}t$, $t=3$. For A:

$\mathbf{v}=(-\mathbf{i}+\mathbf{j})+3(2\mathbf{i}-4\mathbf{j})$

$=(-1+6)\mathbf{i}+(1-12)\mathbf{j}$

$=5\mathbf{i}-11\mathbf{j}$.

$|\mathbf{v}|=\sqrt{5^2+11^2}$

$=\sqrt{25+121}=\sqrt{146}$

$=12.1\,\mathrm{m\,s^{-1}}$ (3 s.f.).

For B:

$\mathbf{v}=\mathbf{i}+3(2\mathbf{j})=\mathbf{i}+6\mathbf{j}$

$|\mathbf{v}|=\sqrt{1^2+6^2}$

$=\sqrt{1+36}=\sqrt{37}$

$=6.08\,\mathrm{m\,s^{-1}}$ (3 s.f.).

b

For A, $\mathbf{s}=\mathbf{u}t+\frac{1}{2}\mathbf{a}t^2$:

$\mathbf{s}=3(-\mathbf{i}+\mathbf{j})+\frac{1}{2}(2\mathbf{i}-4\mathbf{j})9$

$=-3\mathbf{i}+3\mathbf{j}+9\mathbf{i}-18\mathbf{j}$

$=6\mathbf{i}-15\mathbf{j}$.

At collision:

$\mathbf{r}=\mathbf{r}_0+\mathbf{s}$

$=(12\mathbf{i}+12\mathbf{j})+(6\mathbf{i}-15\mathbf{j})$

$=18\mathbf{i}-3\mathbf{j}\,\mathrm{m}$.

c

First find B's displacement:

$\mathbf{s}=\mathbf{u}t+\frac{1}{2}\mathbf{a}t^2$

$=3\mathbf{i}+\frac{1}{2}(2\mathbf{j})9$

$=3\mathbf{i}+9\mathbf{j}$.

Initial position = final position − displacement:

$\mathbf{r}_0=\mathbf{r}-\mathbf{s}$

$=(18\mathbf{i}-3\mathbf{j})-(3\mathbf{i}+9\mathbf{j})$

$=15\mathbf{i}-12\mathbf{j}\,\mathrm{m}$.

Exercise 8A — Y2 Mechanics Pages 162–164 Solution 10

12. A ship is at O at 12:00, with velocity $(-4\mathbf{i}+8\mathbf{j})\,\mathrm{km\,h^{-1}}$. At 14:00 its velocity is $(-2\mathbf{i}-6\mathbf{j})\,\mathrm{km\,h^{-1}}$. At t hours after 12:00, $t\geq 0$, its displacement is $\mathbf{s}$. Model constant acceleration. Find (a) acceleration; (b) $\mathbf{s}(t)$; (c) when directly south-west of O. A second ship has position $\mathbf{r}=(40-25t)\mathbf{j}$ km. (d) Find where they meet.

a

$\mathbf{u}=-4\mathbf{i}+8\mathbf{j}$, $\mathbf{v}=-2\mathbf{i}-6\mathbf{j}$, $t=2$.

$\mathbf{v}=\mathbf{u}+\mathbf{a}t$

$-2\mathbf{i}-6\mathbf{j}=-4\mathbf{i}+8\mathbf{j}+2\mathbf{a}$

$2\mathbf{a}=2\mathbf{i}-14\mathbf{j}$

$\mathbf{a}=\mathbf{i}-7\mathbf{j}$ $\mathrm{km\,h^{-2}}$.

b

$\mathbf{s}=\mathbf{u}t+\frac{1}{2}\mathbf{a}t^2$

$=(-4\mathbf{i}+8\mathbf{j})t+\frac{1}{2}(\mathbf{i}-7\mathbf{j})t^2$

$=(-4t+0.5t^2)\mathbf{i}+(8t-3.5t^2)\mathbf{j}\,\mathrm{km}$.

c

South-west means equal negative components:

$-4t+0.5t^2=8t-3.5t^2$

$4t^2=12t \quad\Rightarrow\quad t=0$ or 3.

Reject $t=0$, at O. At $t=3$, both components are -7.5. This is 3 hours after 12:00, at 15:00.

d

At meeting, $\mathbf{r}=\mathbf{s}$. As $\mathbf{r}$ has no $\mathbf{i}$ component:

$-4t+0.5t^2=0\Rightarrow 0.5t^2=4t\Rightarrow t=0$ or 8.

At $t=0$, positions differ: $0\neq 40\mathbf{j}$. At $t=8$:

$\mathbf{r}=(40-25\times 8)\mathbf{j}=-160\mathbf{j}$,

$8(8)-3.5(8^2)=-160$.

They meet 160 km south of O, at position $-160\mathbf{j}$ km.

CHALLENGE — Y2 Mechanics Page 164 — Solution 11

At an air show, an aeroplane passes over a control tower with velocity $(20\mathbf{i} - 100\mathbf{j})\,\mathrm{m\,s^{-1}}$, flying horizontally with constant acceleration $6\mathbf{j}\,\mathrm{m\,s^{-2}}$. A second plane passes the tower $t > 0$ seconds later, with velocity $(70\mathbf{i} + 40\mathbf{j})\,\mathrm{m\,s^{-1}}$, in a higher horizontal plane with acceleration $-8\mathbf{j}\,\mathrm{m\,s^{-2}}$. They subsequently pass directly over one another. Find t.

Working

Let the horizontal position at crossing be $\mathbf{r}$, and let T be time since the first plane passed the tower.

For the first plane:

$$\mathbf{u} = \begin{pmatrix} 20 \\ -100 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}.$$

$$\mathbf{r} = \mathbf{u}T + \tfrac{1}{2}\mathbf{a}T^2$$

$$= \begin{pmatrix} 20 \\ -100 \end{pmatrix} T + \frac{T^2}{2}\begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$= \begin{pmatrix} 20T \\ -100T + 3T^2 \end{pmatrix}.$$

For the second plane, elapsed time is $T - t$:

$$\mathbf{r} = \begin{pmatrix} 70 \\ 40 \end{pmatrix}(T - t) + \frac{(T-t)^2}{2}\begin{pmatrix} 0 \\ -8 \end{pmatrix}$$

$$= \begin{pmatrix} 70T - 70t \\ 40T - 40t - 4T^2 - 4t^2 + 8Tt \end{pmatrix}.$$

Working

Equate east components:

$$20T = 70T - 70t$$

$$70T - 20T = 70t \quad \Rightarrow \quad T = \frac{7}{5}t.$$

Equate north components:

$$-100T + 3T^2 = 40T - 40t - 4T^2 - 4t^2 + 8Tt$$

$$7T^2 + 4t^2 - 8Tt = 140T - 40t.$$

Substitute $T = 7t/5$:

$$\frac{7 \times 7^2}{25}t^2 + 4t^2 - \frac{8 \times 7}{5}t^2$$

$$= \frac{140 \times 7}{5}t - 40t.$$

$$\left(\frac{343}{25} + 4 - \frac{56}{5}\right)t^2 = (196 - 40)t$$

$$\frac{163}{25}t^2 = 156t.$$

As $t > 0$:

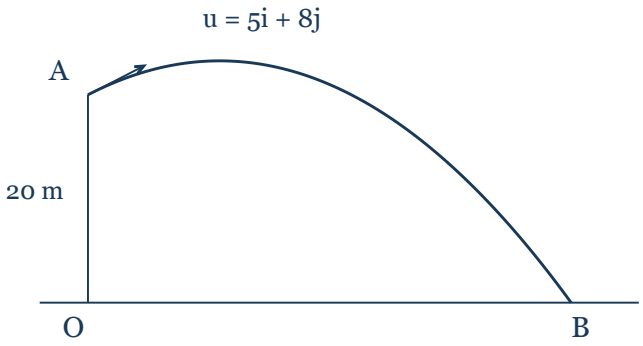
$$t = \frac{3900}{163} = 23.926\ldots\ \mathrm{s}$$

$$= 24\,\mathrm{s} \quad (2\ \text{s.f.}).$$

Vector Methods for Projectiles — Solution 12

A velocity vector gives both direction and magnitude, so can represent projection and subsequent motion.

A racket strikes a ball at A, position $20\mathbf{j}\,\mathrm{m}$ relative to O. Initial velocity is $(5\mathbf{i} + 8\mathbf{j})\,\mathrm{m\,s^{-1}}$, with $\mathbf{i}$ horizontal and $\mathbf{j}$ upward. It moves freely under gravity until hitting horizontal ground at B. Find (a) speed after 1.5 seconds; (b) position $\mathbf{r}(t)$ relative to O; (c) OB.



a

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$= \begin{pmatrix} 5 \\ 8 \end{pmatrix} + \begin{pmatrix} 0 \\ -9.8 \end{pmatrix} t$$

$$= \begin{pmatrix} 5 \\ 8 - 9.8t \end{pmatrix}.$$

At $t = 1.5$:

$$\mathbf{v} = \begin{pmatrix} 5 \\ -6.7 \end{pmatrix} \mathrm{m\,s^{-1}}.$$

$$|\mathbf{v}| = \sqrt{5^2 + 6.7^2}$$

$$= 8.4\,\mathrm{m\,s^{-1}}$$

(2 s.f.).

b

Displacement relative to A:

$$\mathbf{r}_A = \mathbf{u}t + \tfrac{1}{2}\mathbf{a}t^2$$

$$= \begin{pmatrix} 5 \\ 8 \end{pmatrix} t + \tfrac{1}{2}\begin{pmatrix} 0 \\ -9.8 \end{pmatrix} t^2$$

$$= \begin{pmatrix} 5t \\ 8t - 4.9t^2 \end{pmatrix}.$$

Position relative to O:

$$\mathbf{r} = \begin{pmatrix} 0 \\ 20 \end{pmatrix} + \begin{pmatrix} 5t \\ 8t - 4.9t^2 \end{pmatrix}$$

$$= \begin{pmatrix} 5t \\ 8t - 4.9t^2 + 20 \end{pmatrix} \mathrm{m}.$$

c

At ground level, north/up component is zero:

$$8t - 4.9t^2 + 20 = 0.$$

Positive root $t = 2.995\ldots$ s.

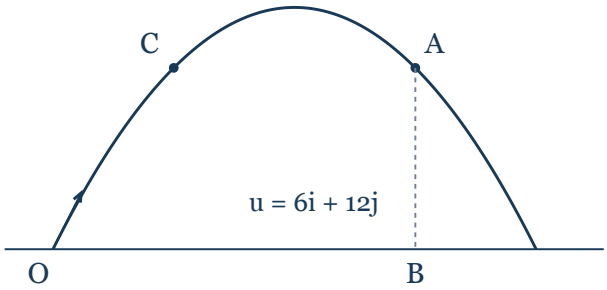
Horizontal component gives

$$OB = 5t = 15\,\mathrm{m} \quad (2\ \text{s.f.}).$$

Test Your Understanding — Solution 13

Edexcel M2 (Old), January 2012, Q7

$\mathbf{i}, \mathbf{j}$ are horizontal and vertical upward unit vectors. A ball is projected from O on horizontal ground with velocity $(6\mathbf{i} + 12\mathbf{j})\,\mathrm{m\,s^{-1}}$. It passes A at time t; B is vertically below A on the ground, with $OB = 2AB$. Find (a) t; (b) speed V at A. At another point C, speed is also V. (c) Find time from O to C. (7,5,3 marks)



a

Horizontal distance: $6t$. Vertical height: $12t - \frac{1}{2}gt^2$.

$2(12t - \frac{1}{2}gt^2) = 6t$

$(24 - 6)t = gt^2$.

As $t > 0$:

$18 = gt \quad \Rightarrow \quad t = \dfrac{18}{g} = 1.84\,\mathrm{s}.$

b

Horizontal velocity is 6. At A, vertical velocity:

$12 - gt = 12 - 18 = -6.$

Thus

$V = \sqrt{6^2 + (-6)^2} = \sqrt{72} = 6\sqrt{2}$

$= 8.49\,\mathrm{m\,s^{-1}}$.

c

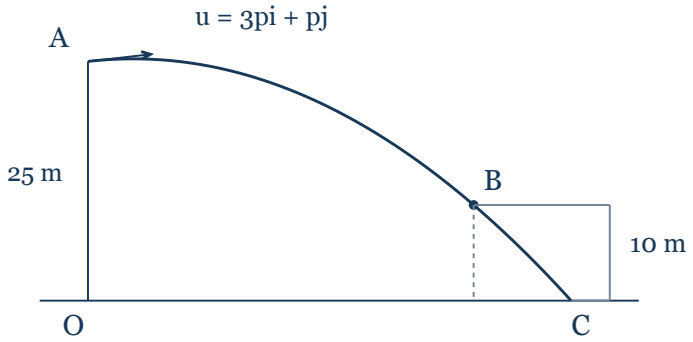
At the other point with this speed, vertical velocity is $+6$:

$12 - gt = 6$

$\Rightarrow t = \dfrac{6}{g} = 0.61\,\mathrm{s}.$

Exercise 8B — Y2 Mechanics Pages 166–167 — Solution 14

3. A ball is projected from A atop a cliff, position $25\mathbf{j}$ m relative to base O at sea level. Initial velocity is $(3p\mathbf{i} + p\mathbf{j})\,\mathrm{m\,s^{-1}}$, with constant p. After 2 seconds it passes B, position $(q\mathbf{i} + 10\mathbf{j})$ m, then hits the sea at C. Model the ball as a particle moving freely under gravity and the sea as horizontal. Find velocity at B.



Working

$\mathbf{u} = 3p\mathbf{i} + p\mathbf{j},\ \mathbf{a} = -9.8\mathbf{j},\ t = 2.$

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$= \begin{pmatrix} 3p \\ p \end{pmatrix} + 2\begin{pmatrix} 0 \\ -9.8 \end{pmatrix} = \begin{pmatrix} 3p \\ p - 19.6 \end{pmatrix}. \quad (1)$

Also, $\mathbf{r}_0 = 25\mathbf{j}$, $\mathbf{r} = q\mathbf{i} + 10\mathbf{j}$:

$\mathbf{s} = \mathbf{r} - \mathbf{r}_0$

$= q\mathbf{i} + 10\mathbf{j} - 25\mathbf{j} = q\mathbf{i} - 15\mathbf{j}.$

Working

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$\begin{pmatrix} q \\ -15 \end{pmatrix} = 2\begin{pmatrix} 3p \\ p \end{pmatrix} + \dfrac{2^2}{2}\begin{pmatrix} 0 \\ -9.8 \end{pmatrix}.$

Compare vertical components:

$-15 = 2p - 19.6$

$2p = 19.6 - 15 \quad \Rightarrow \quad p = \dfrac{4.6}{2} = 2.3.$

Substitute in (1):

$\mathbf{v} = \begin{pmatrix} 3 \times 2.3 \\ 2.3 - 19.6 \end{pmatrix} = \begin{pmatrix} 6.9 \\ -17.3 \end{pmatrix}.$

Velocity at B is $(6.9\mathbf{i} - 17\mathbf{j})\,\mathrm{m\,s^{-1}}$, components to 2 s.f.

Exercise 8B — Y2 Mechanics Pages 166–167 Solution 15

4. A particle P is projected from O on horizontal ground with velocity $(3u\mathbf{i} + 4u\mathbf{j})\,\mathrm{m\,s^{-1}}$. It lands 750 m from O. (a) Show $u = 17.5$. (b) Find greatest height. (c) Find the angle its direction of motion makes with $\mathbf{i}$ when $t = 5\,\mathrm{s}$.

a

$\mathbf{u} = 3u\mathbf{i} + 4u\mathbf{j},\ \mathbf{a} = -9.8\mathbf{j},\ \mathbf{s} = 750\mathbf{i}.$

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$750\mathbf{i} = (3u\mathbf{i} + 4u\mathbf{j})t + \frac{1}{2}(-9.8\mathbf{j})t^2$

$= 3ut\mathbf{i} + (4ut - 4.9t^2)\mathbf{j}.$

Horizontal components:

$750 = 3ut \Rightarrow t = \dfrac{250}{u}.$

Vertical components:

$0 = 4ut - 4.9t^2$

$0 = \dfrac{4u \times 250}{u} - 4.9\left(\dfrac{250}{u}\right)^2$

$= 1000 - \dfrac{306250}{u^2}$

$u^2 = \dfrac{306250}{1000} = 306.25$

$u = \sqrt{306.25} = 17.5.$

b

At greatest height, vertical velocity is zero.

Vertically, $u_y = 4u = 70$, $a = -9.8$, $v_y = 0$.

$v_y^2 = u_y^2 + 2as$

$0 = 70^2 - 2(9.8)s$

$s = \dfrac{70^2}{2 \times 9.8} = 250\,\mathrm{m}.$

c

At $t = 5$,

$\mathbf{v} = \mathbf{u} + \mathbf{a}t$

$= (52.5\mathbf{i} + 70\mathbf{j}) - 5(9.8\mathbf{j})$

$= 52.5\mathbf{i} + 21\mathbf{j}.$

$\tan\theta = \dfrac{v_y}{v_x} = \dfrac{21}{52.5} = 0.4$

$\theta = 21.8^\circ \approx 22^\circ \quad (2\text{ s.f.}),$

above the horizontal.

Exercise 8B — Y2 Mechanics Pages 166–167 Solution 16

5. A particle is projected from O atop a cliff with velocity $(8\mathbf{i} + 10\mathbf{j})\,\mathrm{m\,s^{-1}}$. Moving freely under gravity, it strikes the sea at S after 6 seconds. Find (a) horizontal distance OS; (b) vertical distance OS. At time T, velocity is $(8\mathbf{i} - 14.5\mathbf{j})\,\mathrm{m\,s^{-1}}$. (c) Find T and position relative to O.

a

Let position of S be $x\mathbf{i} + y\mathbf{j}$. Horizontal motion:

$x = u_x t = 8 \times 6 = 48\,\mathrm{m}.$

b

Vertical motion:

$y = u_y t + \frac{1}{2}at^2$

$= 10 \times 6 - 4.9 \times 6^2 = -116.4\,\mathrm{m}.$

The vertical distance is 120 m (2 s.f.), downwards.

c

$\mathbf{v} = \mathbf{u} + \mathbf{a}T$

$8\mathbf{i} - 14.5\mathbf{j} = (8\mathbf{i} + 10\mathbf{j}) - 9.8T\mathbf{j}.$

Compare vertical components:

$-14.5 = 10 - 9.8T$

$T = \dfrac{24.5}{9.8} = \dfrac{5}{2} = 2.5\,\mathrm{s}.$

Then

$\mathbf{s} = \mathbf{u}T + \frac{1}{2}\mathbf{a}T^2$

$= \frac{5}{2}(8\mathbf{i} + 10\mathbf{j}) + \frac{1}{2}(-9.8\mathbf{j})(\frac{5}{2})^2$

$= (8 \times \tfrac{5}{2})\mathbf{i} + \left(10 \times \tfrac{5}{2} - 4.9(\tfrac{5}{2})^2\right)\mathbf{j}$

$= \left(20\mathbf{i} - \dfrac{45}{8}\mathbf{j}\right)\ \mathrm{m}.$

Exercise 8B — Y2 Mechanics Pages 166–167 Solution 17

6. Use $g = 10\,\text{m s}^{-2}$. A body B is projected from O on horizontal ground with velocity $(a\mathbf{i} + b\mathbf{j})\,\text{m s}^{-1}$, where $a, b > 0$. It moves freely under gravity until landing at P, immediately coming to rest. Its position on the path is $(x\mathbf{i} + y\mathbf{j})\,\text{m}$.

(a) Show $y = bx/a - 5x^2/a^2$.

(b) Given $a = 8$, range $OP = X$ and maximum height Y, find (i) X, (ii) Y, in terms of b.

a

$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

$x\mathbf{i} + y\mathbf{j} = (a\mathbf{i} + b\mathbf{j})t + \frac{1}{2}(-10\mathbf{j})t^2.$

Horizontal components:

$x = at \Rightarrow t = \dfrac{x}{a}. \quad (1)$

Vertical components:

$y = bt - 5t^2. \quad (2)$

Substitute (1) in (2):

$y = \dfrac{bx}{a} - \dfrac{5x^2}{a^2},$

as required.

b.i

At landing, $x = X, y = 0, a = 8$:

$0 = \dfrac{bX}{8} - \dfrac{5X^2}{64}$

$\dfrac{5X^2}{64} = \dfrac{bX}{8} \Rightarrow 5X^2 = 8bX.$

Reject $X = 0$:

$X = \dfrac{8b}{5} = 1.6b.$

b.ii

Maximum height occurs halfway along the range:

$x = \dfrac{X}{2} = \dfrac{4b}{5}.$

$Y = \dfrac{b \times 4b}{8 \times 5} - \dfrac{5(4b)^2}{64 \times 5^2}$

$= \dfrac{b^2}{10} - \dfrac{b^2}{20} = \dfrac{b^2}{20} = 0.05b^2.$

Variable Acceleration in One Dimension Solution 18

$s \xrightarrow{d/dt} v \xrightarrow{d/dt} a$

$s \xleftarrow{\int v\,dt} v \xleftarrow{\int a\,dt} a$

In Mechanics Year 1, velocity is rate of change of displacement and acceleration is rate of change of velocity:

$v = \dfrac{ds}{dt}, \qquad a = \dfrac{dv}{dt}.$

Reverse by integration: $v = \int a\,dt$, $s = \int v\,dt$, with constants found from initial conditions. Stay in one dimension, now using more complex Pure Year 2 functions.

A particle moving in a straight line has

$a = \cos(2\pi t)\,\text{m s}^{-2}, \quad t \geq 0.$

At $t = 0$, $v = 1/(2\pi)\,\text{m s}^{-1}$. Find (a) $v(t)$; (b) maximum speed; (c) distance travelled in the first 3 seconds.

a

$v = \int \cos(2\pi t)\,dt$

$= \dfrac{1}{2\pi}\sin(2\pi t) + c.$

Reverse chain rule: divide by the constant multiplying t. At $t = 0$:

$v = 0 + c = \dfrac{1}{2\pi}.$

$v = \dfrac{1}{2\pi}\big(\sin(2\pi t) + 1\big).$

b

Maximum sine value is 1:

$v_{\max} = \dfrac{1}{2\pi} + \dfrac{1}{2\pi} = \dfrac{1}{\pi}\,\text{m s}^{-1}.$

Velocity is nonnegative, so this is also maximum speed.

c

Area under the velocity–time graph gives distance because $v \geq 0$. Factor out $1/(2\pi)$:

$s = \dfrac{1}{2\pi}\int_0^3 (\sin(2\pi t) + 1)\,dt$

$= \dfrac{1}{2\pi}\left[-\dfrac{1}{2\pi}\cos(2\pi t) + t\right]_0^3$

$= \dfrac{3}{2\pi} = 0.477\,\text{m} \quad (3\text{ s.f.}).$

Test Your Understanding — Solution 19

A particle of mass 6 kg moves on the positive x-axis. Its displacement from the origin is

$$s = 2t^{3/2} + \frac{e^{-2t}}{3}\,\text{m}, \quad t \geq 0.$$

(a) Find velocity when $t = 1.5$. A single force of magnitude F acts along positive x. (b) Find F when $t = 2$.

a

$$v = \frac{ds}{dt} = 3t^{1/2} - \frac{2}{3}e^{-2t}\,\text{m s}^{-1}.$$

At $t = 1.5$:

$$v = 3(1.5)^{0.5} - \frac{2}{3}e^{-3}$$

$$= 3.64\,\text{m s}^{-1} \quad (3\text{ s.f.}).$$

b

$$a = \frac{dv}{dt} = 1.5t^{-0.5} + \frac{4}{3}e^{-2t}.$$

At $t = 2$:

$$a = 1.5(2)^{-0.5} + \frac{4}{3}e^{-4}$$

$$= 1.0850\ldots\ \text{m s}^{-2}.$$

$$F = ma = 6(1.0850\ldots) = 6.51\,\text{N}.$$

Chain-rule recap:

$$\frac{d}{dx}(e^{kx}) = ke^{kx}.$$

Exercise 8C — Y2 Mechanics Pages 168–170 — Solution 20

7. A particle moves in a straight line with acceleration $3\sqrt{t}\,\text{m s}^{-2}$, $t > 0$, measured from when it begins motion. After 1 second, velocity is 2 m/s. Find the time taken to travel 16 m.

Working

Integrate twice to obtain displacement, then find when 16 m has been travelled.

$$v = \int a\,dt = \int 3\sqrt{t}\,dt$$

$$= 3 \times \frac{2}{3}t^{3/2} + c = 2t^{3/2} + c.$$

Use $t = 1, v = 2$:

$$2 = 2(1^{3/2}) + c \Rightarrow c = 0.$$

$$v = 2t^{3/2}.$$

Working

$$s = \int v\,dt = \int 2t^{3/2}\,dt$$

$$= \frac{2}{5} \times 2t^{5/2} + c_2 = \frac{4}{5}t^{5/2} + c_2.$$

Measure displacement from the starting position, so $s(0) = 0$ gives $c_2 = 0$. Since $v \geq 0$, distance equals displacement:

$$16 = \frac{4}{5}t^{5/2}$$

$$20 = t^{5/2} \Rightarrow t = 20^{2/5} = 3.3144\ldots.$$

It takes 3.31 s (3 s.f.).

Exercise 8C — Y2 Mechanics Pages 168–170 Solution 21

8. In a 200 m straight sprint, a runner's displacement from the start is modelled by $s = k\sqrt{t}$, $0 \le t \le T$, in metres and seconds. She finishes in 25 seconds. Find (a) k, T; (b) her speed at the finish; (c) a criticism for small t.

a

At the finish $s = 200, t = 25$, so $T = 25\,\text{s}$.

$$200 = k\sqrt{25} \quad \Rightarrow \quad k = \frac{200}{5} = 40.$$

b

$$v = \frac{ds}{dt} = \tfrac{1}{2}(40)t^{-1/2} = \frac{20}{\sqrt{t}}.$$

$$v(25) = \frac{20}{\sqrt{25}} = 4\,\text{m s}^{-1}.$$

c

For small positive t, predicted speed is unrealistically large. For example, at $t = 0.01\,\text{s}$:

$v = 20(0.01)^{-1/2} = 200\,\text{m s}^{-1}$.

No human can run this fast; the model is unsuitable near the start.

Exercise 8C — Y2 Mechanics Pages 168–170 Solution 22

9. A particle moves in a straight line with velocity $v = 2 + 8\sin(kt)\,\text{m s}^{-1}$, $t \ge 0$. Initial acceleration is $4\,\text{m s}^{-2}$. (a) Find k. Using it, (b) find all $t \in [0, 4\pi]$ when $a = 0$; (c) show $4a^2 = 64 - (v-2)^2$; (d) find maximum velocity and maximum acceleration.

a

$$a = \frac{dv}{dt} = 8k\cos(kt).$$

At $t = 0, a = 4$:

$$4 = 8k \quad \Rightarrow \quad k = \tfrac{1}{2}.$$

b

$$a = 8 \times \tfrac{1}{2}\cos(\tfrac{1}{2}t) = 4\cos(\tfrac{1}{2}t).$$

For $a = 0$:

$$\cos(\tfrac{1}{2}t) = 0$$

$$\tfrac{1}{2}t = \tfrac{\pi}{2}, \tfrac{3\pi}{2} \Rightarrow t = \pi, 3\pi.$$

c

$$64 - (v-2)^2 = 64 - (8\sin(\tfrac{1}{2}t))^2$$

$$= 64 - 64\sin^2(\tfrac{1}{2}t)$$

$$= 64(1 - \sin^2(\tfrac{1}{2}t))$$

$$= 64\cos^2(\tfrac{1}{2}t)$$

$$= 4(4\cos(\tfrac{1}{2}t))^2 = 4a^2.$$

d

At a maximum of velocity, $a = 0$. The candidates from (b) are $\pi, 3\pi$:

$\sin(\pi/2) = 1$,

$\sin(3\pi/2) = -1$.

Thus $v_{\max} = 2 + 8 = 10\,\text{m s}^{-1}$; the other is a minimum.

At an interior acceleration extremum, $da/dt = 0$. Since a is a multiple of cosine, its derivative is a multiple of sine. Thus $\sin(t/2) = 0$, so $v = 2$. From (c):

$$4a^2 = 64 - (2-2)^2 = 64$$

$$a = \pm 4.$$

The maximum is $+4\,\text{m s}^{-2}$, attained when cosine is 1 (including $t = 0$).

Exercise 8C — Y2 Mechanics Pages 168–170 — Solution 23

10. A particle moves along the x-axis, with velocity in the positive direction

$$v = \begin{cases} 10t - 2t^{3/2}, & 0 \le t \le 4, \\ 24 - \left(\frac{t-4}{2}\right)^4, & t > 4. \end{cases}$$

At $t = 0$, it is at the origin. Find (a) greatest speed on $[0, 4]$; (b) distance from O at $t = 4$; (c) when at rest for $t > 4$; (d) total distance in the first 10 seconds.

a

$a = \dfrac{dv}{dt} = \dfrac{d}{dt}(10t - 2t^{3/2})$

$= 10 - 3\sqrt{t}.$

For $0 \le t \le 4$, $a \ge 0$ and $v \ge 0$, so speed increases. Its maximum is

$v(4) = 10(4) - 2(4^{3/2})$

$= 24\,\text{m s}^{-1}.$

b

For the first 4 seconds:

$s = \int (10t - 2t^{3/2})\,dt$

$= 5t^2 - \dfrac{4}{5}t^{5/2} + c.$

At $t = 0$, $s = 0$, hence $c = 0$.

$s(4) = 5(4^2) - \dfrac{4}{5}(4^{5/2})$

$= 80 - \dfrac{128}{5} = 54.4\,\text{m}.$

c

At rest, $v = 0$:

$0 = 24 - \left(\dfrac{t-4}{2}\right)^4.$

As $t > 4$,

$\dfrac{t-4}{2} = \sqrt[4]{24}$

$t = 4 + 2\sqrt[4]{24} = 8.4267\ldots\,\text{s}$

$= 8.43\,\text{s} \quad (3\text{ s.f.}).$

d

Let $T = 4 + 2\sqrt[4]{24}$. First 4 seconds: 54.4 m. Split at reversal:

$D_{4,10} = \left|\int_4^T \left(24 - \left(\dfrac{t-4}{2}\right)^4\right) dt\right|$

$+ \left|\int_T^{10} \left(24 - \left(\dfrac{t-4}{2}\right)^4\right) dt\right|.$

Use $G(t) = 24t - (t-4)^5/80$.

$G(T) = 180.99317\ldots,$

$G(4) = 96,\ G(10) = 142.8.$

$D_{4,10} = |84.99317\ldots|$

$+ |-38.19317\ldots|$

$= 123.18634\ldots\,\text{m} \approx 123.2\,\text{m}.$

Total distance:

$54.4 + 123.18634\ldots$

$= 177.58634\ldots\,\text{m} \approx 177.6\,\text{m}.$

Differentiating Vectors — Solution 24

If $\mathbf{v} = \begin{pmatrix} t^2 \\ \sin t \end{pmatrix}$, what is the acceleration?

Derivative

Differentiate the two components independently:

$\mathbf{a} = \dfrac{d\mathbf{v}}{dt} = \begin{pmatrix} 2t \\ \cos t \end{pmatrix}.$

If $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$:

$\mathbf{v} = \dfrac{d\mathbf{r}}{dt} = \dot{\mathbf{r}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j},$

$\mathbf{a} = \dfrac{d\mathbf{v}}{dt} = \dfrac{d^2\mathbf{r}}{dt^2} = \ddot{\mathbf{r}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}.$

Dot notation means differentiation with respect to time, e.g. $\dot{x} = dx/dt$; it is common in Physics.

A particle P, mass 0.8 kg, is acted on by a single force $\mathbf{F}$. Its position is

$\mathbf{r} = 2t^3\mathbf{i} + 50t^{-1/2}\mathbf{j}\,\text{m}, \quad t > 0.$

Find (a) speed at $t = 4$; (b) acceleration vector at $t = 2$; (c) force at $t = 2$.

a

$\mathbf{v} = \dot{\mathbf{r}} = \begin{pmatrix} 6t^2 \\ -25t^{-3/2} \end{pmatrix}.$

At $t = 4$:

$\mathbf{v} = \begin{pmatrix} 96 \\ -25/8 \end{pmatrix} \text{m s}^{-1}.$

$|\mathbf{v}| = \sqrt{96^2 + (25/8)^2}$

$= 96.1\,\text{m s}^{-1} \quad (3\text{ s.f.}).$

b

$\mathbf{a} = \dot{\mathbf{v}} = \begin{pmatrix} 12t \\ (75/2)t^{-5/2} \end{pmatrix}.$

At $t = 2$:

$\mathbf{a} = \begin{pmatrix} 24 \\ 6.6291\ldots \end{pmatrix} \text{m s}^{-2}.$

c

$\mathbf{F} = m\mathbf{a} = 0.8\begin{pmatrix} 24 \\ 6.6291\ldots \end{pmatrix}$

$= \begin{pmatrix} 19.2 \\ 5.30 \end{pmatrix} \text{N}.$

4. Position relative to fixed O is

$\mathbf{r} = 4t^2\mathbf{i} + (24t - 3t^2)\mathbf{j}\,\text{m}, \quad t \geq 0.$

(a) Find speed at $t = 2$. (b) Show acceleration is constant and find its magnitude.

a

$\mathbf{v} = \dot{\mathbf{r}} = 8t\mathbf{i} + (24 - 6t)\mathbf{j}.$

At $t = 2$:

$\mathbf{v} = 16\mathbf{i} + 12\mathbf{j}$

$|\mathbf{v}|^2 = 16^2 + 12^2 = 400$

$|\mathbf{v}| = \sqrt{400} = 20\,\text{m s}^{-1}.$

b

$\mathbf{a} = \dot{\mathbf{v}} = 8\mathbf{i} - 6\mathbf{j}.$

Neither component depends on t, so acceleration is constant.

$|\mathbf{a}|^2 = 8^2 + (-6)^2 = 100$

$|\mathbf{a}| = \sqrt{100} = 10\,\text{m s}^{-2}.$

5. Particle P, mass 0.5 kg, starts at O, projected at $t = 0$. Its position is

$\mathbf{r} = (t^3 - 12t)\mathbf{i} + (4t^2 - 6t)\mathbf{j}\,\text{m}, \quad t \geq 0.$

Find (a) projection speed; (b) when moving parallel to $\mathbf{j}$; (c) position then. A single variable force $\mathbf{F}$ causes the motion. (d) Find $|\mathbf{F}|$ at $t = 5$.

a

$\mathbf{v} = \dot{\mathbf{r}} = (3t^2 - 12)\mathbf{i} + (8t - 6)\mathbf{j}.$

At $t = 0$:

$\mathbf{v} = -12\mathbf{i} - 6\mathbf{j}$

$|\mathbf{v}|^2 = (-12)^2 + (-6)^2 = 180$

$|\mathbf{v}| = \sqrt{180} = 6\sqrt{5}\,\text{m s}^{-1}.$

b

Motion parallel to $\mathbf{j}$ means zero $\mathbf{i}$ component of velocity:

$3t^2 - 12 = 0 \Rightarrow t^2 = 4$

$\Rightarrow t = 2 \quad (t \geq 0).$

c

At $t = 2$:

$\mathbf{r} = (2^3 - 12 \times 2)\mathbf{i}$

$+ (4 \times 2^2 - 6 \times 2)\mathbf{j}$

$= (-16\mathbf{i} + 4\mathbf{j})\,\text{m}.$

d

$\mathbf{v} = (3t^2 - 12)\mathbf{i} + (8t - 6)\mathbf{j}$

$\mathbf{a} = \dot{\mathbf{v}} = 6t\mathbf{i} + 8\mathbf{j}.$

At $t = 5$:

$\mathbf{a} = 30\mathbf{i} + 8\mathbf{j}.$

$\mathbf{F} = m\mathbf{a} = 0.5(30\mathbf{i} + 8\mathbf{j})$

$= 15\mathbf{i} + 4\mathbf{j}\,\text{N}.$

$|\mathbf{F}| = \sqrt{15^2 + 4^2} = 15.524\ldots$

$= 15.5\,\text{N} \quad (3\text{ s.f.}).$

Exercise 8D — Y2 Mechanics Pages 171–173 Solution 27

6. A particle moving in a plane has position

$\mathbf{r} = (3t^2 - 6t + 4)\mathbf{i} + (t^3 + kt^2)\mathbf{j}$ m.

At $t = 3$, its speed is $12\sqrt{5}\,\text{m s}^{-1}$. (a) Find both possible k. (b) For each, find acceleration magnitude at $t = 1.5$.

a

$\mathbf{v} = (6t - 6)\mathbf{i} + (3t^2 + 2kt)\mathbf{j}$.

At $t = 3$:

$\mathbf{v} = 12\mathbf{i} + (27 + 6k)\mathbf{j}$.

$(12\sqrt{5})^2 = |\mathbf{v}|^2$

$720 = 12^2 + (27 + 6k)^2$

$720 = 144 + 729 + 324k + 36k^2$

$0 = 36k^2 + 324k + 153$

Divide by 9:

$0 = 4k^2 + 36k + 17$

$0 = (2k + 1)(2k + 17)$

$k = -0.5 \quad \text{or} \quad -8.5$.

b

If $k = -0.5$:

$\mathbf{v} = (6t - 6)\mathbf{i} + (3t^2 - t)\mathbf{j}$

$\mathbf{a} = 6\mathbf{i} + (6t - 1)\mathbf{j}$.

At $t = 1.5$:

$\mathbf{a} = 6\mathbf{i} + 8\mathbf{j}$

$|\mathbf{a}|^2 = 6^2 + 8^2 = 100$

$\Rightarrow |\mathbf{a}| = 10$.

b

If $k = -8.5$:

$\mathbf{v} = (6t - 6)\mathbf{i} + (3t^2 - 17t)\mathbf{j}$

$\mathbf{a} = 6\mathbf{i} + (6t - 17)\mathbf{j}$.

At $t = 1.5$:

$\mathbf{a} = 6\mathbf{i} - 8\mathbf{j}$

$|\mathbf{a}|^2 = 6^2 + (-8)^2 = 100$

$\Rightarrow |\mathbf{a}| = 10$.

For both values, magnitude is $10\,\text{m s}^{-2}$.

Exercise 8D — Y2 Mechanics Pages 171–173 Solution 28

7. A particle's position is

$\mathbf{r} = 6t^2\mathbf{i} + t^{5/2}\mathbf{j}$ m, $\quad t \geq 0$.

At $t = 4$, find (a) speed; (b) acceleration as a vector.

a

$\mathbf{v} = \dot{\mathbf{r}} = 12t\mathbf{i} + \frac{5}{2}t^{3/2}\mathbf{j}$.

At $t = 4$, using $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8$:

$\mathbf{v} = 48\mathbf{i} + \frac{5}{2}(4^{3/2})\mathbf{j}$

$= 48\mathbf{i} + 20\mathbf{j}$.

$|\mathbf{v}|^2 = 48^2 + 20^2 = 2704$

$|\mathbf{v}| = \sqrt{2704} = 52\,\text{m s}^{-1}$.

b

Remember $\mathbf{a} = \dot{\mathbf{v}} = \ddot{\mathbf{r}}$.

$\mathbf{a} = 12\mathbf{i} + \frac{5}{2}\frac{3}{2}t^{1/2}\mathbf{j}$

$= 12\mathbf{i} + \frac{15}{4}t^{1/2}\mathbf{j}$.

At $t = 4$:

$\mathbf{a} = 12\mathbf{i} + \frac{15}{4}\sqrt{4}\mathbf{j}$

$= \left(12\mathbf{i} + \frac{15}{2}\mathbf{j}\right)\,\text{m s}^{-2}$.

Exercise 8D — Y2 Mechanics Pages 171–173 Solution 29

8. A particle moves in a horizontal plane with

$\mathbf{r} = (18t - 4t^3)\mathbf{i} + ct^2\mathbf{j}\,\text{m}, \quad t \geq 0,\ c > 0.$

At $t = 1.5$, speed is 15 m/s. Find (a) c; (b) acceleration at $t = 1.5$.

8a

$\mathbf{v} = (18 - 12t^2)\mathbf{i} + 2ct\mathbf{j}.$

At $t = 1.5$:

$\mathbf{v} = (18 - 12 \times 1.5^2)\mathbf{i} + 3c\mathbf{j}$

$= -9\mathbf{i} + 3c\mathbf{j}.$

$15^2 = |\mathbf{v}|^2 = (-9)^2 + (3c)^2$

$9c^2 = 144.$

As $c > 0$, $c = 4$.

8b

$\mathbf{a} = \dot{\mathbf{v}} = -24t\mathbf{i} + 2c\mathbf{j}.$

With $c = 4, t = 1.5$:

$\mathbf{a} = (-36\mathbf{i} + 8\mathbf{j})\,\text{m s}^{-2}.$

9. A particle has position

$\mathbf{r} = (2t^2 - 3t)\mathbf{i} + (5t + t^2)\mathbf{j}\,\text{m}, \quad t \geq 0.$

Show acceleration is constant and find its magnitude.

9

$\mathbf{v} = (4t - 3)\mathbf{i} + (5 + 2t)\mathbf{j}$

$\mathbf{a} = 4\mathbf{i} + 2\mathbf{j}.$

It is constant because it contains no t.

$|\mathbf{a}| = \sqrt{4^2 + 2^2} = 2\sqrt{5}\,\text{m s}^{-2}.$

Exercise 8D — Y2 Mechanics Pages 171–173 Solution 30

10. A particle moves in a horizontal plane with

$\mathbf{r} = (20t - 2t^3)\mathbf{i} + kt^2\mathbf{j}\,\text{m}, \quad t \geq 0,\ k > 0.$

At $t = 2$, speed is 16 m/s. Find (a) k; (b) acceleration when moving parallel to $\mathbf{j}$.

a

$\mathbf{v} = \dot{\mathbf{r}} = (20 - 6t^2)\mathbf{i} + 2kt\mathbf{j}.$

$\mathbf{v}(2) = (20 - 24)\mathbf{i} + 4k\mathbf{j}$

$= -4\mathbf{i} + 4k\mathbf{j}.$

$16^2 = |\mathbf{v}(2)|^2 = (-4)^2 + (4k)^2$

$256 = 16 + 16k^2$

$k^2 = \dfrac{256 - 16}{16} = 15.$

Since $k > 0$, $k = \sqrt{15}$.

b

Parallel to $\mathbf{j}$ means zero $\mathbf{i}$ component of velocity:

$20 - 6t^2 = 0$

$t^2 = \dfrac{20}{6} \Rightarrow t = \sqrt{\dfrac{10}{3}}.$

Now

$\mathbf{a} = \dot{\mathbf{v}} = -12t\mathbf{i} + 2\sqrt{15}\mathbf{j}.$

At this time:

$\mathbf{a} = -12\sqrt{\dfrac{10}{3}}\mathbf{i} + 2\sqrt{15}\mathbf{j}$

$= (-4\sqrt{30}\mathbf{i} + 2\sqrt{15}\mathbf{j})\,\text{m s}^{-2}.$

Integrating Vectors

Solution 31

Integrate vector components independently. A particle moving in a plane has velocity

$\mathbf{v} = 3t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} \quad \text{m s}^{-1}, \quad t \geq 0.$

At $t = 0$, position relative to O is $(2\mathbf{i} - 3\mathbf{j})$ m. Find its position vector at time t.

Integrate

$$\mathbf{r} = \int \mathbf{v}\, dt = \int \begin{pmatrix} 3t \\ t^2/2 \end{pmatrix} dt$$

$$= \begin{pmatrix} 3t^2/2 \\ t^3/6 \end{pmatrix} + \mathbf{c}.$$

The constant of integration is a vector.

Position

Use the initial position to find the constant:

$$t = 0: \quad \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \mathbf{c}.$$

$$\mathbf{c} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}.$$

$$\boxed{\mathbf{r} = \begin{pmatrix} 3t^2/2 + 2 \\ t^3/6 - 3 \end{pmatrix} \text{m}.}$$

Further Example

Solution 32

A particle has acceleration $\mathbf{a} = 4\mathbf{i} - 2t\mathbf{j}\,\text{m s}^{-2}$. At $t = 3$, velocity is $6\mathbf{i}\,\text{m s}^{-1}$ and position is $(20\mathbf{i} + 3\mathbf{j})$ m. Find (a) direction of motion at $t = 2$; (b) distance from O at $t = 0$.

a

$$\mathbf{v} = \int \mathbf{a}\, dt = \begin{pmatrix} 4t \\ -t^2 \end{pmatrix} + \mathbf{c}.$$

Use $t = 3$, not zero, for the given velocity:

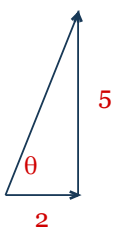
$$\begin{pmatrix} 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 12 \\ -9 \end{pmatrix} + \mathbf{c} \Rightarrow \mathbf{c} = \begin{pmatrix} -6 \\ 9 \end{pmatrix}.$$

$$\mathbf{v} = \begin{pmatrix} 4t - 6 \\ -t^2 + 9 \end{pmatrix}.$$

At $t = 2$, $\mathbf{v} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$.

$$\tan\theta = \frac{5}{2} \Rightarrow \theta = 68.2^\circ$$

above the positive $\mathbf{i}$ direction.



b

$$\mathbf{r} = \int \mathbf{v}\, dt = \begin{pmatrix} 2t^2 - 6t \\ -t^3/3 + 9t \end{pmatrix} + \mathbf{d}.$$

Use the position at $t = 3$:

$$\begin{pmatrix} 20 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 18 \end{pmatrix} + \mathbf{d} \Rightarrow \mathbf{d} = \begin{pmatrix} 20 \\ -15 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 2t^2 - 6t + 20 \\ -t^3/3 + 9t - 15 \end{pmatrix}.$$

Initially $\mathbf{r} = \begin{pmatrix} 20 \\ -15 \end{pmatrix}$, so

$$OP = \sqrt{20^2 + (-15)^2} = \sqrt{625} = 25\,\text{m}.$$

Test Your Understanding — Solution 33

Edexcel M2 (Old), January 2013, Q4

Particle P has velocity $(4t-5)\mathbf{i}+3\mathbf{j}\,\text{m s}^{-1}$, and initial position $(2\mathbf{i}+5\mathbf{j})$ m. Find (a) when moving parallel to $\mathbf{j}$; (b) its position at time t. Particle Q has constant velocity $(-2\mathbf{i}+c\mathbf{j})\,\text{m s}^{-1}$, initial position $(11\mathbf{i}+2\mathbf{j})$ m. They collide at $(d\mathbf{i}+14\mathbf{j})$ m. Find (c)(i) c, (ii) d.

a

$4t-5=0 \Rightarrow t=\frac{5}{4}$ s.

b

$\mathbf{r}_P=(2t^2-5t)\mathbf{i}+3t\mathbf{j}+\mathbf{C}$.

At $t=0$, $\mathbf{C}=2\mathbf{i}+5\mathbf{j}$.

$\mathbf{r}_P=(2t^2-5t+2)\mathbf{i}+(3t+5)\mathbf{j}$.

c

$\mathbf{r}_Q=(11-2t)\mathbf{i}+(2+ct)\mathbf{j}$.

At collision, $3t+5=14$, so $t=3$.

$2+3c=14 \Rightarrow c=4$.

$d=11-2(3)=5$.

Equivalently, $d=2(3)^2-5(3)+2=5$.

Alternative

Alternative: equate east components:

$2t^2-5t+2=11-2t$

$2t^2-3t-9=(2t+3)(t-3)=0$.

Reject negative time, giving $t=3$.

Or use $t=(11-d)/2$:

$$2\left(\frac{11-d}{2}\right)^2-5\left(\frac{11-d}{2}\right)+2=d$$

$d^2-19d+70=0$

$(d-5)(d-14)=0$.

Reject $d=14$, which gives negative time; $d=5$.

Exercise 8E — Integrating Vectors — Solution 34

1. A particle has acceleration

$\mathbf{a}=6t^2\mathbf{i}+(8-4t^3)\mathbf{j}\,\text{m s}^{-2}$.

It is initially at rest at O. Find (a) velocity at $t=2$; (b) position at $t=4$.

a

$\mathbf{v}=\int \mathbf{a}\,dt=2t^3\mathbf{i}+(8t-t^4)\mathbf{j}+\mathbf{C}$.

At $t=0$, $\mathbf{v}=\mathbf{0}$, so $\mathbf{C}=\mathbf{0}$.

$\mathbf{v}(2)=16\mathbf{i}+(16-16)\mathbf{j}$

$=16\mathbf{i}\,\text{m s}^{-1}$.

b

$\mathbf{r}=\int \mathbf{v}\,dt=\frac{1}{2}t^4\mathbf{i}+\left(4t^2-\frac{1}{5}t^5\right)\mathbf{j}+\mathbf{D}$.

At $t=0$, $\mathbf{r}=\mathbf{0}$, so $\mathbf{D}=\mathbf{0}$.

$\mathbf{r}(4)=128\mathbf{i}+(64-204.8)\mathbf{j}$

$=(128\mathbf{i}-140.8\mathbf{j})$ m.

Exercise 8E — Integrating Vectors

Solution 35

2. A particle has velocity

$\mathbf{v} = (3t^2 + 2)\mathbf{i} + (6t - 4)\mathbf{j}\,\text{m s}^{-1}, \quad t \geq 0.$

At $t = 2$, its position is $9\mathbf{j}$ m. Find (a) its initial distance from O; (b) its acceleration when moving parallel to $\mathbf{i}$.

a

$\mathbf{r} = (t^3 + 2t)\mathbf{i} + (3t^2 - 4t)\mathbf{j} + \mathbf{A}.$

At $t = 2$:

$9\mathbf{j} = 12\mathbf{i} + 4\mathbf{j} + \mathbf{A}$

$\mathbf{A} = -12\mathbf{i} + 5\mathbf{j}.$

$\mathbf{r} = (t^3 + 2t - 12)\mathbf{i} + (3t^2 - 4t + 5)\mathbf{j}.$

Initially $\mathbf{r} = -12\mathbf{i} + 5\mathbf{j}.$

$|\mathbf{r}|^2 = (-12)^2 + 5^2 = 169$

$|\mathbf{r}| = 13\,\text{m}.$

b

Parallel to $\mathbf{i}$:

$6t - 4 = 0 \Rightarrow t = \frac{2}{3}.$

$\mathbf{a} = \dfrac{d\mathbf{v}}{dt} = 6t\mathbf{i} + 6\mathbf{j}.$

Hence

$\mathbf{a} = 4\mathbf{i} + 6\mathbf{j}\,\text{m s}^{-2}.$

Exercise 8E — Integrating Vectors

Solution 36

3. A particle has acceleration $\mathbf{a} = (2t - 4)\mathbf{i} + 6\sin t\,\mathbf{j}\,\text{m s}^{-2}$. It is at rest when $t = \pi/2$. Find (a) velocity at time t; (b) exact speed at $t = 3\pi/2$.

a

$\mathbf{v} = (t^2 - 4t)\mathbf{i} - 6\cos t\,\mathbf{j} + \mathbf{c}.$

At $t = \pi/2$:

$\mathbf{0} = (\pi^2/4 - 2\pi)\mathbf{i} + 0\mathbf{j} + \mathbf{c}$

$\mathbf{c} = (2\pi - \pi^2/4)\mathbf{i}.$

$\mathbf{v} = (t^2 - 4t + 2\pi - \pi^2/4)\mathbf{i} - 6\cos t\,\mathbf{j}.$

b

At $t = 3\pi/2$:

$\mathbf{v} = (9\pi^2/4 - 6\pi + 2\pi - \pi^2/4)\mathbf{i} - 0\mathbf{j}$

$= (8\pi^2/4 - 6\pi + 2\pi)\mathbf{i}$

$= (2\pi^2 - 4\pi)\mathbf{i}.$

As $2\pi^2 - 4\pi > 0$, the speed is

$2\pi^2 - 4\pi\,\text{m s}^{-1}.$

Exercise 8E — Integrating Vectors

Solution 37

4. A particle has acceleration $\mathbf{a} = (5t - 3)\mathbf{i} + (8 - t)\mathbf{j}\,\text{m s}^{-2}$ and initial velocity $(2\mathbf{i} - 5\mathbf{j})\,\text{m s}^{-1}$. Find (a) velocity at time t; (b) when moving parallel to $\mathbf{i} - \mathbf{j}$; (c) its speed then.

a

$\mathbf{v} = (\frac{5}{2}t^2 - 3t)\mathbf{i}$
$+ (8t - \frac{1}{2}t^2)\mathbf{j} + \mathbf{C}.$
Initially $\mathbf{C} = 2\mathbf{i} - 5\mathbf{j}$, so
$\mathbf{v} = (\frac{5}{2}t^2 - 3t + 2)\mathbf{i}$
$+ (8t - \frac{1}{2}t^2 - 5)\mathbf{j}.$

b

The components must be opposites:
$\frac{5}{2}t^2 - 3t + 2 = -(8t - \frac{1}{2}t^2 - 5)$
$2t^2 + 5t - 3 = 0$
$(2t - 1)(t + 3) = 0.$
Reject $t = -3$, leaving $t = \frac{1}{2}$ s.

c

At $t = \frac{1}{2}$:
$\mathbf{v} = (\frac{5}{8} - \frac{3}{2} + 2)\mathbf{i} + (4 - \frac{1}{8} - 5)\mathbf{j}$
$= \frac{9}{8}\mathbf{i} - \frac{9}{8}\mathbf{j}.$
$|\mathbf{v}|^2 = (\frac{9}{8})^2 + (-\frac{9}{8})^2$
$|\mathbf{v}| = \dfrac{9\sqrt{2}}{8}\,\text{m s}^{-1}.$

Exercise 8E — Integrating Vectors

Solution 38

5. Particle P has acceleration $2\mathbf{i} - 2t\mathbf{j}\,\text{m s}^{-2}$, initial velocity $2\mathbf{j}\,\text{m s}^{-1}$ and initial position $6\mathbf{i}$ m relative to fixed O. (a) Find its position at time t. Particle Q has velocity $(3t^2 - 4)\mathbf{i} - 2t\mathbf{j}\,\text{m s}^{-1}$. They collide at $t = 3$. (b) Find Q's initial position.

a

$\mathbf{v}_P = 2t\mathbf{i} - t^2\mathbf{j} + \mathbf{A}.$
Initially $\mathbf{A} = 2\mathbf{j}$, hence
$\mathbf{v}_P = 2t\mathbf{i} + (2 - t^2)\mathbf{j}.$
$\mathbf{r}_P = t^2\mathbf{i} + (2t - t^3/3)\mathbf{j} + \mathbf{B}.$
Initially $\mathbf{B} = 6\mathbf{i}$, so
$\mathbf{r}_P = (t^2 + 6)\mathbf{i} + (2t - t^3/3)\mathbf{j}.$

b

$\mathbf{r}_Q = (t^3 - 4t)\mathbf{i} - t^2\mathbf{j} + \mathbf{C}.$
At $t = 3$:
$\mathbf{r}_P = (9 + 6)\mathbf{i} + (6 - 9)\mathbf{j} = 15\mathbf{i} - 3\mathbf{j}.$
Since they collide:
$15\mathbf{i} - 3\mathbf{j} = (27 - 12)\mathbf{i} - 9\mathbf{j} + \mathbf{C}$
$\mathbf{C} = 6\mathbf{j}.$
$\mathbf{r}_Q = (t^3 - 4t)\mathbf{i} + (6 - t^2)\mathbf{j}.$
Initial position is $6\mathbf{j}$ m.

Exercise 8E — Integrating Vectors

Solution 39

6. A particle starts from rest at position $(4\mathbf{i} - 6\mathbf{j})$ m. Its acceleration is $(4t - 3)\mathbf{i} - 6t^2\mathbf{j}\,\mathrm{m\,s^{-2}}$. Find (a) velocity at $t = 1/2$; (b) position at $t = 6$.

a

$\mathbf{v} = (2t^2 - 3t)\mathbf{i} - 2t^3\mathbf{j} + \mathbf{A}.$

Initially at rest, so $\mathbf{A} = \mathbf{0}$. At $t = 1/2$:

$\mathbf{v} = (2(\tfrac{1}{2})^2 - \tfrac{3}{2})\mathbf{i} - 2(\tfrac{1}{2})^3\mathbf{j}$

$= (-\mathbf{i} - \tfrac{1}{4}\mathbf{j})\,\mathrm{m\,s^{-1}}.$

b

$\mathbf{r} = (\tfrac{2}{3}t^3 - \tfrac{3}{2}t^2)\mathbf{i} - \tfrac{1}{2}t^4\mathbf{j} + \mathbf{B}.$

Initially $\mathbf{B} = 4\mathbf{i} - 6\mathbf{j}$, so

$\mathbf{r} = (\tfrac{2}{3}t^3 - \tfrac{3}{2}t^2 + 4)\mathbf{i} - (\tfrac{1}{2}t^4 + 6)\mathbf{j}.$

At $t = 6$:

$\mathbf{r} = (144 - 54 + 4)\mathbf{i} - (648 + 6)\mathbf{j}$

$= (94\mathbf{i} - 654\mathbf{j})\,\mathrm{m}.$

Exercise 8E — Integrating Vectors

Solution 40

8. Particle P has velocity $(4t - 3)\mathbf{i} + 4\mathbf{j}\,\mathrm{m\,s^{-1}}$ and initial position $(\mathbf{i} + 2\mathbf{j})$ m. (a) Find its position at time t. Particle Q has velocity $(5\mathbf{i} + k\mathbf{j})\,\mathrm{m\,s^{-1}}$ and initial position $(11\mathbf{i} + 5\mathbf{j})$ m. They collide. Find (b)(i) k, (ii) the collision position.

a

$\mathbf{r}_P = (2t^2 - 3t)\mathbf{i} + 4t\mathbf{j} + \mathbf{c}.$

Initially $\mathbf{c} = \mathbf{i} + 2\mathbf{j}$, hence

$\mathbf{r}_P = (2t^2 - 3t + 1)\mathbf{i} + (4t + 2)\mathbf{j}.$

b.i

$\mathbf{r}_Q = \int (5\mathbf{i} + k\mathbf{j})\,dt$

$= 5t\mathbf{i} + kt\mathbf{j} + \mathbf{c}.$

At $t = 0$:

$11\mathbf{i} + 5\mathbf{j} = \mathbf{0} + \mathbf{c},$

$\mathbf{c} = 11\mathbf{i} + 5\mathbf{j}.$

Hence

$\mathbf{r}_Q = (5t + 11)\mathbf{i} + (kt + 5)\mathbf{j}.$

At collision, east components are equal:

$2t^2 - 3t + 1 = 5t + 11$

$2t^2 - 8t - 10 = 0$

$t^2 - 4t - 5 = (t - 5)(t + 1) = 0.$

Reject $t = -1$: $t = 5$. North components give

$20 + 2 = 5k + 5$

$k = \dfrac{22 - 5}{5} = 3.4.$

b.ii

$\mathbf{r}_Q(5) = (25 + 11)\mathbf{i}$

$+ (5 \times 3.4 + 5)\mathbf{j}$

$= (36\mathbf{i} + 22\mathbf{j})\,\mathrm{m}.$